

NGC Variable	Legacy Variable	Usage
#14021- #14026	N/A	G110 (G154 P2) additional work offsets
#14041- #14386	N/A	G110 (G154 P3- G154 P20) additional work offsets
#14401- #14406	N/A	G110 (G154 P21) additional work offsets
#14421- #15966	N/A	G110 (G154 P22- G154 P99) additional work offsets
#20000- #29999	N/A	Setting
#30000- #39999	N/A	Parameter
#32014	N/A	Machine Serial Number
#50001- #50200	N/A	Tool Type
#50201- #50400	N/A	Tool material
#50401- #50600	N/A	Tool Offset Point
#50601- #50800	N/A	Estimated RPM
#50801- #51000	N/A	Estimated Feedrate
#51001- #51200	N/A	Offset Pitch
#51201- #51400	N/A	Actually VPS Estimated RPM
#51401- #51600	N/A	Work material
#51601- #51800	N/A	VPS Feedrate
#51801- #52000	N/A	X Approximate Probe Length
#52001- #52200	N/A	Y Approximate Probe Length
#52201- #52400	N/A	Z Approximate Probe Length
#52401- #52600	N/A	Approximate Probe Diameter
#52601- #52800	N/A	Edge Measure Height
#52801- #53000	N/A	Tool Tolerance
#53201- #53400	N/A	Probe Type

NGC Variable	Legacy Variable	Usage
#53401- #53600	N/A	Live Tool Radius
#53601- #53800	N/A	Live Tool Radius Wear
#53801- #54000	N/A	X Geometry
#54001- #54200	N/A	Y Geometry
#54201- #54400	N/A	Z Geometry
#54401- #54600	N/A	Diameter Geometry
#54601- #54800	N/A	Tip
#54801- #55000	N/A	X Geometry Wear
#55001- #55200	N/A	Y Geometry Wear
#55201- #55400	N/A	Z Geometry Wear
#55401- #55600	N/A	Diameter Wear

6.7.3 System Variables In-Depth

System variables are associated with specific functions. A detailed description of these functions follows.

#550-#699 #10550- #10699 General and Probe Calibration Data

These general purpose variables are saved on power off. Some of these higher #5xx variables store probe calibration data. Example: #592 sets which side of the table the tool probe is positioned. If these variables are overwritten, you will need to calibrate the probe again.



NOTE:

If the machine does not have a probe installed, you can use these variables as general-purpose variables saved on power off.

#1080-#1097 #11000-#11255 #13000-#13063 1-Bit Discrete Inputs

You can connect designated inputs from external devices with these macros:

Variables	Legacy Variables	Usage
#11000-#11255		256 discrete inputs (read only)
#13000-#13063	#1080-#1087 #1090-#1097	Raw and Filtered analog to digital inputs (read only)

Specific input values can be read from within a program. The format is #11nnn where nnn is the Input Number. Press **[DIAGNOSTIC]** and select the I/O tab to see the Input and Output numbers for different devices.

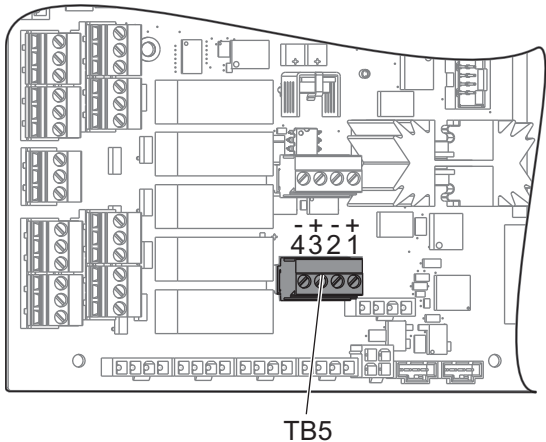
Example:

#10000=#11018

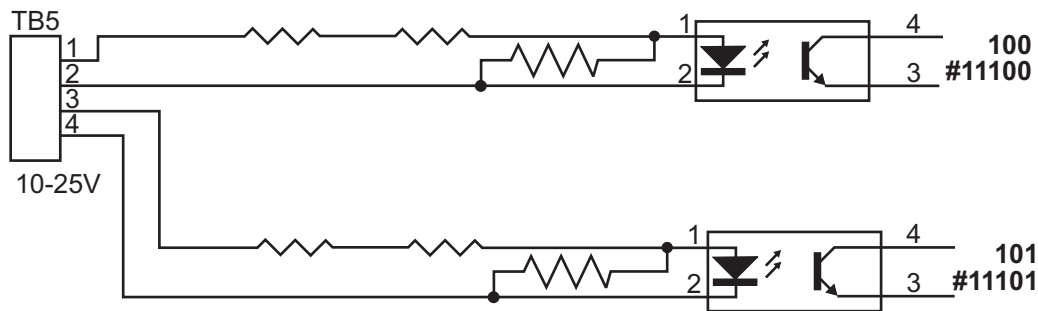
This example records the state of #11018, which refers to Input 18 (M-Fin_Input), to variable #10000.

User Inputs on the I/O PCB

The I/O PCB includes a set of (2) available inputs (100 (#11100) and 101 (#11101)) at TB5.



Devices connected to these inputs must have their own power supply. When a device applies 10-25 V between Pins 1 and 2, the input 100 bit (Macro #11100) changes from 1 to 0. When a device applies 10-25 V between pins 3 and 4, the input 101 (Macro #11101) bit changes from 1 to 0.



#12000-#12255 1-Bit Discrete Outputs

The Haas control is capable of controlling up to 256 discrete outputs. However, a number of these outputs are reserved for the Haas control to use.

Variables	Legacy Variables	Usage
#12000-#12255		256 discrete outputs

Specific output values can be read, or written to, from within a program. The format is #12nnn where nnn is the Output Number.

Example:

#10000=#12018 ;

This example records the state of #12018, which refers to Input 18 (Coolant Pump Motor), to variable #10000.

#1064-#1268 Maximum Axis Loads

These variables contain the maximum load an axis has achieved since the machine was last powered on, or since that Macro Variable was cleared. The Maximum Axis Load is the greatest load (100.0 = 100%) an axis has seen, not the Axis Load at the time that the control reads the variable.

#1064 = X Axis	#1264 = C axis
#1065 = Y Axis	#1265 = U axis

#1066 = Z Axis	#1266 = V axis
#1067 = A Axis	#1267 = W axis
#1068 = B Axis	#1268 = T axis

Tool Offsets

Use these following macro variables to read or set the following geometry, shift, or wear offset values:

#2001-#2050	X-Axis geometry/shift offset
#2051-#2100	Y-Axis geometry/shift offset
#2101-#2150	Z-Axis geometry/shift offset
#2201-#2250	Tool nose radius geometry
#2301-#2350	Tool tip direction
#2701-#2750	X-Axis tool wear
#2751-#2800	Y-Axis tool wear
#2801-#2850	Z-Axis tool wear
#2901-#2950	Tool nose radius wear

#3000 Programmable Alarm Messages

#3000 Alarms can be programmed. A programmable alarm will act like the built-in alarms. An alarm is generated by setting macro variable #3000 to a number between 1 and 999.

```
#3000= 15 (MESSAGE PLACED INTO ALARM LIST) ;
```

When this is done, *Alarm* flashes at the bottom of the display and the text in the next comment is placed into the alarm list. The alarm number (in this example, 15) is added to 1000 and used as an alarm number. If an alarm is generated in this manner all motion stops and the program must be reset to continue. Programmable alarms are always numbered between 1000 and 1999.

#3001-#3002 Timers

Two timers can be set to a value by assigning a number to the respective variable. A program can then read the variable and determine the time passed since the timer was set. Timers can be used to imitate dwell cycles, determine part-to-part time or wherever time-dependent behavior is desired.

- #3001 Millisecond Timer - The millisecond timer represents the system time after power on in number of milliseconds. The whole number returned after accessing #3001 represents the number of milliseconds.
- #3002 Hour Timer - The hour timer is similar to the millisecond timer except that the number returned after accessing #3002 is in hours. The hour and millisecond timers are independent of each other and can be set separately.

#3003 Single Block Suppression

Variable #3003 overrides the Single Block function in G-code. When #3003 has a value of 1, the control executes each G-code command continuously even though the Single Block function is ON. When #3003 has a value of zero, Single Block operates as normal. You must press **[CYCLE START]** to execute each line of code in single block mode.

```
#3003=1 ;  
G54 G00 X0 Z0 ;  
G81 R0.2 Z-0.1 F.002 L0 ;  
S2000 M03 ;  
#3003=0 ;  
T02 M06 ;  
Q.05 G83 R0.2 Z-1. F.001 L0 ;  
X0. Z0. ;  
...
```

#3004 Enables and Disables Feed Hold

Variable #3004 overrides specific control features during operation.

The first bit disables **[FEED HOLD]**. If variable #3004 is set to 1, **[FEED HOLD]** is disabled for the program blocks that follow. Set #3004 to 0 to enable **[FEED HOLD]** again. For example:

```
...  
(Approach code - [FEED HOLD] allowed) ;  
#3004=1 (Disables [FEED HOLD]) ;  
(Non-stoppable code - [FEED HOLD] not allowed) ;  
#3004=0 (Enables [FEED HOLD]) ;  
(Depart code - [FEED HOLD] allowed) ;  
...
```

This is a map of variable #3004 bits and the associated overrides.

E = Enabled D = Disabled

#3004	Feed Hold	Feed Rate Override	Exact Stop Check
0	E	E	E
1	D	E	E
2	E	D	E
3	D	D	E
4	E	E	D
5	D	E	D
6	E	D	D
7	D	D	D



NOTE:

When the feed rate override variable is set (#3004 = 2), the control will set the feed rate override to 100% (default). During #3004 = 2 the control will show 100% in red bold text on the display until the variable is reset. Once the feed rate override is reset (#3004 = 0) the feed rate will be restored to the previous value before setting the variable.

#3006 Programmable Stop

You can add stops to the program that act like an M00 - The control stops and waits until you press **[CYCLE START]**, then the program continues with the block after the #3006. In this example, the control displays the comment on the lower-center of the screen.

```
#3006=1 (comment here) ;
```

#3030 Single Block

In Next Generation Control when the system variable #3030 is set to a 1; the control will go into single block mode. There is no need to limit the lookahead using a G103 P1, the Next Generation Control will correctly process this code.

**NOTE:**

For the Classic Haas Control to process system variable #3030=1 correctly, it is necessary to limit the lookahead to 1 block using a G103 P1 before the #3030=1 code.

#4001-#4021 Last Block (Modal) Group Codes

G-code groups let the machine control process the codes more efficiently. G-codes with similar functions are usually in the same group. For example, G90 and G91 are under group 3. Macro variables #4001 through #4021 store the last or default G code for any of 21 groups.

G-Codes Group number is listed next to it's description in the G-Code section.

Example:

G81 Drill Canned Cycle (Group 09)

When a macro program reads the group code, the program can change the behavior of the G-code. If #4003 contains 91, then a macro program could determine that all moves should be incremental rather than absolute. There is no associated variable for group zero; group zero G codes are Non-modal.

#4101-#4126 Last Block (Modal) Address Data

Address codes A-Z (excluding G) are maintained as modal values. The information represented by the last line of code interpreted by the lookahead process is contained in variables #4101 through #4126. The numeric mapping of variable numbers to alphabetic addresses corresponds to the mapping under alphabetic addresses. For example, the value of the previously interpreted D address is found in #4107 and the last interpreted I value is #4104. When aliasing a macro to an M-code, you may not pass variables to the macro using variables #1 - #33. Instead, use the values from #4101 - #4126 in the macro.

#5001-#5006 Last Target Position

The final programmed point for the last motion block can be accessed through variables #5001 - #5006, X, Z, Y, A, B, and C respectively. Values are given in the current work coordinate system and can be used while the machine is in motion.

#5021-#5026 Current Machine Coordinate Position

#5021 X-Axis	#5022 Z Axis	#5023 Y-Axis
#5024 A-Axis	#5025 B-Axis	#5026 C-axis

To get the current machine axis positions, call macro variables #5021- #5025 corresponding to axis X, Z, Y, A, and B, respectively.


NOTE:

Values CANNOT be read while the machine is in motion.

#5041-#5046 Current Work Coordinate Position

To get the current machine axis positions, call macro variables #5041-#5046 corresponding to axis X, Z, Y, A, B, and C, respectively.


NOTE:

The values CANNOT be read while the machine is in motion.

The value of #504X has tool length compensation applied to it.

#5061-#5069 Current Skip Signal Position

Macro variables #5061-#5069 corresponding to X, Z, Y, A, B, C, U, V and W respectively, give the axis positions where the last skip signal occurred. Values are given in the current work coordinate system and can be used while the machine is in motion.

The value of #5062 (Z) has tool length compensation applied to it.

#5081-#5086 Tool Length Compensation

Macro variables #5081 - #5086 give the current total tool length compensation in axis X, Z, Y, A, B, or C, respectively. This includes tool length offset referenced by the current value set in T plus the wear value.

#5201-#5326, #7001-#7386, #14001-#14386 Work Offsets

Macro expressions can read and set all work offsets. This lets you preset coordinates to exact locations, or set coordinates to values based upon the results of skip signal (probed) locations and calculations. When any of the offsets are read, the interpretation look-ahead queue is stopped until that block is executed.

#5201- #5206	G52 X, Z, Y, A, B, C offset values
#5221- #5226	G54 X, Z, Y, A, B, C offset values
#5241- #5246	G55 X, Z, Y, A, B, C offset values
#5261- #5266	G56 X, Z, Y, A, B, C offset values

#5281- #5286	G57 X, Z, Y, A, B, C offset values
#5301- #5306	G58 X, Z, Y, A, B, C offset values
#5321- #5326	G59 X, Z, Y, A, B, C offset values
#7001- #7006	G110 (G154 P1) additional work offsets
#7021-#7026 (#14021-#14026)	G111 (G154 P2) additional work offsets
#7041-#7046 (#14041-#14046)	G114 (G154 P3) additional work offsets
#7061-#7066 (#14061-#14066)	G115 (G154 P4) additional work offsets
#7081-#7086 (#14081-#14086)	G116 (G154 P5) additional work offsets
#7101-#7106 (#14101-#14106)	G117 (G154 P6) additional work offsets
#7121-#7126 (#14121-#14126)	G118 (G154 P7) additional work offsets
#7141-#7146 (#14141-#14146)	G119 (G154 P8) additional work offsets
#7161-#7166 (#14161-#14166)	G120 (G154 P9) additional work offsets
#7181-#7186 (#14181-#14186)	G121 (G154 P10) additional work offsets
#7201-#7206 (#14201-#14206)	G122 (G154 P11) additional work offsets
#7221-#7226 (#14221-#14221)	G123 (G154 P12) additional work offsets
#7241-#7246 (#14241-#14246)	G124 (G154 P13) additional work offsets
#7261-#7266 (#14261-#14266)	G125 (G154 P14) additional work offsets

#7281-#7286 (#14281-#14286)	G126 (G154 P15) additional work offsets
#7301-#7306 (#14301-#14306)	G127 (G154 P16) additional work offsets
#7321-#7326 (#14321-#14326)	G128 (G154 P17) additional work offsets
#7341-#7346 (#14341-#14346)	G129 (G154 P18) additional work offsets
#7361-#7366 (#14361-#14366)	G154 P19 additional work offsets
#7381-#7386 (#14381-#14386)	G154 P20 additional work offsets

#6001-#6250 Settings Access with Macro Variables

Access settings through variables #20000 - #20999 or #6001 - #6250, starting from setting 1 respectively. Refer to page **365** for the detailed descriptions of the settings that are available in the control.



NOTE:

The #20000 - 20999 range numbers correspond directly to Setting numbers. You should use #6001 - #6250 for settings access only if you need your program to be compatible with older Haas machines.

#6198 Next-Generation Control Identifier

The macro variable #6198 has a read-only value of 1000000.

You can test #6198 in a program to detect the control version, and then conditionally run program code for that control version. For example:

```
%
```

```
IF[#6198 EQ 1000000] GOTO5 ;
```

```
(Non-NGC code) ;
```

GOTO6 ;

N5 (NGC code) ;

N6 M30 ;

%

In this program, if the value stored in #6198 is equal to 1000000, go to Next Generation Control compatible code then end the program. If if the value stored in #6198 is not equal to 1000000, run the non-NGC program and then end the program.

#7501 - #7806, #3028 Pallet Changer Variables

The status of the pallets from the Automatic Pallet Changer is checked with these variables:

#7501-#7506	Pallet priority
#7601-#7606	Pallet status
#7701-#7706	Part program numbers assigned to pallets
#7801-#7806	Pallet usage count
#3028	Number of pallet loaded on receiver

#8500-#8515 Advanced Tool Management

These variables give information on Advanced Tool Management (ATM). Set variable #8500 to the tool group number, then access information for the selected tool group with the read-only macros #8501-#8515.

#8500	Advanced Tool Management (ATM). Group ID
#8501	ATM. Percent of available tool life of all tools in the group.
#8502	ATM. Total available tool usage count in the group.
#8503	ATM. Total available tool hole count in the group.

#8504	ATM. Total available tool feed time (in seconds) in the group.
#8505	ATM. Total available tool total time (in seconds) in the group.
#8510	ATM. Next tool number to be used.
#8511	ATM. Percent of available tool life of the next tool.
#8512	ATM. Available usage count of the next tool.
#8513	ATM. Available hole count of the next tool.
#8514	ATM. Available feed time of the next tool (in seconds).
#8515	ATM. Available total time of the next tool (in seconds).

#8550-#8567 Advanced Tool Management Tooling

These variables give information on tooling. Set variable #8550 to the tool group number, then access information for the selected tool with the read-only macros #8551-#8567.



NOTE:

Macro variables #1601-#2800 give access to the same data for individual tools as #8550-#8567 give for Tool Group tools.

#8550	Individual tool ID
#8551	Number of Flutes on tool
#8552	Maximum recorded vibration
#8553	Tool length offset
#8554	Tool length wear
#8555	Tool diameter offset
#8556	Tool diameter wear
#8557	Actual diameter
#8558	Programmable coolant position

#8559	Tool feed timer (seconds)
#8560	Total tool timers (seconds)
#8561	Tool life monitor limit
#8562	Tool life monitor counter
#8563	Tool load monitor maximum load sensed so far
#8564	Tool load monitor limit

#50001 - #50200 Tool Type

Use macro variables #50001 - #50200, to read or write the tool type set in the tool offset page.

T6.2: Available Tool Types for Lathe

Tool Type	Tool Type #
OD Turn	21
OD Groove	22
OD Thread	23
Part Off	24
Drill	25
ID Turn	26
ID Groove	27
ID Thread	28
Face Groove	29
Tap	30
Probe	31
Reserve for Future Use	32-40